

Bialetti Gioia CF90 — Disassembly Assumptions

Documentation of the assumptions made for the disassembly use-case modelled with the futurdata-thesis AND/OR editor

Scope of this document

This document records the assumptions underlying the disassembly model `disassembly_bialetti_gioia_v2.json`, which describes the step-by-step disassembly of the Bialetti Gioia CF90 domestic capsule espresso machine for the purpose of repair or end-of-life material recovery. The model follows the grammar required by the supervisor: diamond nodes are composite operations, action circles are sequential atomic sub-steps that explain how to perform the operation, and each diamond produces the extracted parts plus a reduced 'remainder' composite that carries the disassembly forward.

The Bialetti Gioia is documented by its manufacturer through a user-facing instruction manual that intentionally does not include any internal disassembly procedure (the manual explicitly directs internal repairs to authorised service centres). Consequently, the product-level specifications used in the model are taken from the official manual, while the internal procedure, fasteners, connectors and per-component engineering values are modelled by analogy with the typical layout of entry-level capsule espresso machines in the same market segment. This document distinguishes the two categories explicitly.

1. Verified product-level data

The following facts are documented in the Bialetti Gioia CF90 official instruction manual (Bialetti Industrie, n.d.) and can be cited directly. Page references are to the multilanguage PDF version (Italian section 5-31, English section 33-59).

Specification	Value	Source in manual
Dimensions (W x D x H)	93 x 340 x 221 mm	Section "Technical Data"
Voltage	220-240 V, 50-60 Hz	Section "Technical Data"
Rated power	1200 W	Section "Technical Data"
Water tank capacity	0.5 L, removable	Section "Technical Data"
Used-capsule drawer	max 8 capsules	Legend, item 11
Capsule type	Bialetti aluminium only	Introduction
Capsule recognition	Inductive sensor for aluminium	Introduction; Troubleshooting
Stand-by behaviour	Auto after 9 min, < 0.5 W	Section "Energy saving mode"
Heating system	Boiler ("caldaia")	Section "First use"
Programmable dose	Range 20-180 mL	Section "Setting dispensing"
Power-cord servicing	Authorised service only	Warnings

A terminology note is worth flagging: the user manual consistently uses *caldaia* / *boiler* for the heating subsystem, while the commercial product description used in Bialetti's catalogue and retailer spec sheets advertises a *Thermoblock* heating system. The two refer to the same physical subsystem in this product class. The model uses *thermoblock* consistently because it is the technically more precise term for a hot-on-demand block-heater (as opposed to a tank boiler).

2. Inferred internal layout and components

The Bialetti manual does not document the internal architecture beyond the external legend on the technical-data page. The internal-component model below is therefore inferred from the typical architecture of single-boiler capsule espresso machines in the entry-level segment (machines such as Krups Inissia, De Longhi Lattissima Pro and Lavazza A Modo Mio Jolie, which share the same broad mechanical pattern).

Internal subsystems modelled

Subsystem	Justification for inclusion
Vibration pump	Required to produce the 20 bar declared on the product spec. Solenoid vibration pumps (Ulka EX5 / E
Hall-effect flow meter	Inferred from the documented programmable-dose feature ("setting dispensing quantity", manual sect
Silicone tubing set	Required to connect the tank valve, the pump and the heating subsystem. Food-grade silicone with sp
Thermoblock assembly	Documented as "Thermoblock heating system" in the commercial spec. Modelled as cast aluminium b
Main control PCB	Required by the on/off button, the dispense buttons, the LED indicators, the auto-standby logic, the do
Aluminium-capsule sensor	Explicitly documented in the manual ("innovative sensor"). The physical type (inductive coil) is inferred
Front button panel	Documented in the legend (items 4, 5, 6). Modelled as a small PCB with tactile switches and indicator
Wiring harness + power cord	Required for any mains-powered appliance. Connector families (JST-XH for logic-level, JST-PH for se
Outer plastic shell	Documented as "Corpo apparecchio" in the legend (item 10). The "Responsible" variant is advertised
Chassis frame	Required to host all internal subsystems. Modelled as galvanized steel structural elements with ABS b

3. Per-leaf assumptions: material and weight

The model lists, for each leaf component, a material composition and a weight in grams. None of these values is measured on a real unit. They are engineering estimates that satisfy two constraints simultaneously: (a) consistency with the materials commonly used for each subsystem in this appliance class, and (b) the sum of leaf weights staying within the declared total of ~2500 g. The table below records the values and their rationale.

Component	Material (assumed)	Weight (g)	Rationale
Outer shell	ABS (50% PCR on Responsible value)	260	Largest plastic part. PCR content documented by Bialetti for the R
Button panel	FR-4 PCB + switches + LEDs + ABS cover	35	Small front-mounted board, low component count.
Capsule sensor	Copper coil + ferrite core + plastic housing	18	Compact inductive sensor.
Vibration pump	Steel body + copper coil + brass + rubber	165	Typical mass of an Ulka EX5 vibration pump.
Silicone tubing	Food-grade silicone + steel clamps	22	Three short tubes plus clamps.
Flow meter	ABS + ceramic shaft + Hall PCB	28	Compact inline meter.
Main PCB	FR-4 + SMD + electrolytic + triacs	55	Single-board controller for low-cost capsule machines.
Wiring harness	Copper + PVC + nylon JST housing	60	Internal harness covering all subsystems.
Power cord	3-core Cu cable + PVC + Schuko plug	140	Short mains cord (~0.8 m), typical for the form factor.
Chassis frame	Galvanised steel + ABS + EPDM feet	480	Structural frame. Sized to bring the total close to declared mass.
Heating element	NiCr 80/20 wire + ceramic + stainless sheath	45	Compact element rated 1200 W at 230 V.
Safety thermostat	Brass + bimetal disc + ceramic	15	Threaded bimetal capsule.
Aluminium body	Cast AlSi9Cu3 alloy	240	Thermal-mass block of the thermoblock; key CRM-recoverable ma
Stainless tube	AISI 304	60	Thin-wall water-path tube embedded in the aluminium body.

Total of assumed leaf weights: 1653 g. The remaining ~850 g of declared total mass is accounted for by the parts not modelled as separate leaves (structural fasteners, internal brackets, gaskets, the brewing chamber with piercing needles, and packaging-internal components such as the wooden capsule opener included by Bialetti) and by the natural conservative margin of the declared figure.

4. Disassembly procedure: fasteners, connectors and ordering

The disassembly procedure is the part of the model most exposed to uncertainty, because the official manual does not document any internal step. The procedure modelled here reflects the canonical sequence for small household appliances of this class (e.g. as described in Failli & Dini, 2005, on destructive disassembly of household appliances): open the outer shell first, then peel off the subsystems in order of their connectivity to the chassis, leaving the heating subsystem (the most thermally and structurally embedded) for last.

Fastener assumptions

Specific screw types are assumed by analogy with the typical practice of Italian / European small-appliance manufacturers in this price range:

Where	Assumed type	Justification
Bottom plate, 4 screws	Torx T10	Tamper-resistant heads on load-bearing fastenings are standard.
Rear panel, 2 screws	Phillips PH1	Hidden rear screws are commonly Phillips PH1 in this segment.
Button-panel mount, 2 screws	Phillips PH0	Small mounting screws on light plastic parts are typically PH0.
PCB mounting, 4 screws	Phillips PH0	Same as button panel.
Pump bracket, 2 screws	Phillips PH1	Mid-load mount.
Thermoblock bracket, 2 screws	Phillips PH1	Mid-load mount.
Thermoblock outlet	Brass nut, 10 mm	Standard hydraulic fitting size.
Side shell clips	Plastic snap clips	Universal in ABS housings; require a flat plastic pry tool.

Connector assumptions

Low-voltage connector families assumed by analogy with typical practice in entry-level capsule machines:

Connection	Assumed connector	Justification
Button board to PCB	JST-XH, 4-pin	Common pitch for low-current logic signals.
Capsule sensor to PCB	JST-PH, 2-pin	Smaller pitch, typical for sensor leads.
Pump to PCB	JST-XH or similar, 2-pin	Triac output to pump.
Thermostat to PCB	JST-PH, 2-pin	Safety-loop wiring.
Heating element to harness	Faston 6.3 mm	Standard for mains-current heating loads.
Mains input to PCB	Screw-terminal block	L / N / PE on the board edge.

5. Grammar choices in the AND/OR model

The model implements the disassembly grammar specified by the supervisor. Two grammar choices in particular shape the resulting graph and are worth documenting here, because they reflect interpretive decisions that other modellers might have taken differently.

5.1 Action circles in sequential chains

Under each diamond, the action circles that describe how to perform the composite operation are not attached in parallel but in a sequential chain: diamond -> step 1 -> step 2 -> ... -> step n. The graph therefore encodes the ordering explicitly, not only through step naming (D1.1, D1.2, ...). This is deliberately compatible with the downstream guide generator: a generator walking the chain produces a numbered list of instructions automatically.

5.2 Linear chain of reduced composites

When several subsystems are removed in succession from the same chassis, the model does not fan out the chassis into all subsystems at once. Instead, each removal operation produces (a) the extracted part(s) and (b) a reduced composite representing the chassis with one fewer subsystem inside. This yields a linear chain:

```
Internal chassis -> [remove panel] -> Chassis without panel -> [extract hydraulics] ->  
Chassis without panel/hydraulics -> ... -> Bare chassis frame
```

This is the grammar required by the supervisor and matches the physical reality of disassembly: you are working on the same chassis throughout, and that chassis gets lighter and emptier with each step.

5.3 Natural groups vs single-leaf extractions

Where multiple leaf parts form a natural operational group (for example, the vibration pump, its silicone tubing and the flow meter are extracted as a single hydraulic block in one operation), the model uses a single diamond that produces the multiple leaves at once plus the reduced composite. The alternative (one diamond per leaf in a longer chain) would produce a graph more granular than the physical operation; the chosen convention follows the supervisor's instruction that natural groups may be extracted together.

6. References

Bialetti Industrie S.p.A. (n.d.). *Bialetti Gioia CF90 - Istruzioni per l'uso / Instructions for use*. Official multilanguage user manual. Available from https://www.bialetti.com/media/manual/macchine-espresso/IM_CF90-Gioia_IT-EN-FR-DE-ES.pdf

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